

GPE Project: Intermediate Report

Cellular and Wi-Fi Infrastructure Installation - MV World Voyager

Alice Navalho (ist1106873) — Maria Abrantes (ist1106339) — Beatriz Coelho (ist1107113)
Mónica Ferreira (ist1106520) — Luís Nóbrega (ist1106716) — Rodrigo Esgueira (ist1106089)

1 Objectives

The main goal of this project is to design and plan the complete deployment of a dedicated mobile communications network and Wi-Fi infrastructure on the cruise ship MV World Voyager. As the Project Leader, my mission is to ensure a smooth integration of telecommunication services that allow passengers to stay connected via satellite links while the ship is at sea. This needs a practical approach that balances technical possibilities with legal requirements, ensuring the cellular system only operates in international waters to prevent any interference with local operators at port.

From a business viewpoint, the project seeks to improve the ship's appeal by meeting current passenger demands for consistent internet access and quality voice services. This requires close coordination among various internal departments, such as Radio Engineering, Legal, Marketing, and Buying, as well as with external stakeholders like the Ship Owner (Atlas Ocean Voyages), the Constructor, and Satellite Service Providers. The project leader must make sure that all these parties work together to provide a reliable backhaul connection and a strong internal distribution of signals that meet the operator's quality standards.

For project governance and operational management, the aim is to create a solid project management plan using MS Project to track the schedule and manage the critical path. The project is structured through a detailed Work Breakdown Structure (WBS) that oversees every phase, from initial site visits and location negotiations to final integration and testing. Ultimately, the goal is to deliver a fully functional communication system within the set timeframe and budget, while actively identifying and addressing risks related to construction delays, equipment procurement, and international roaming agreements. Additionally, the financial planning will strictly follow the required labor categories of €80/h, €100/h, and €120/h.

2 Stakeholder's Framework

The success of the World Voyager connectivity project depends on the strategic alignment and active participation of a diverse group of internal and external actors. This framework identifies the key stakeholders, ranging from corporate sponsors and specialized engineering departments to external regulatory bodies and the vessel's operational command. By analyzing their specific interests, potential impact, and perceived risks, the project team can develop a targeted communication and engagement strategy. This analysis serves as the primary input for the Power/Interest Matrix, ensuring that resources are prioritized toward managing the stakeholders with the greatest capacity to influence the project's critical path and commercial outcome.

Table 1: Project Stakeholder Analysis & Management Plan

Stakeholder (name and role)	Stakeholder's interest in the project?	Stakeholder's potential impact	What does the project expect the Stakeholder to provide?	Perceived attitudes and/or risk	Inputs to Stakeholder Communication Plan
Project Leader & Project Development Dept.	Successful delivery of the network infrastructure within the 2026 academic deadlines and allocated budget.	Critical/High: Controls the WBS, resource allocation, and daily management of the critical path.	Project Management Plan (MS Project), coordination between all internal/external parties, and technical reporting.	Highly Committed: Goal-oriented. Risk: Potential over-allocation of resources or delays in dependency tasks.	Weekly status meetings, internal progress reports.
Atlas Ocean Voyages (Customer / Ship Owner)	Provide premium connectivity to passengers to stay competitive; ensure ship safety and aesthetic integrity.	High: Can grant or deny access to the vessel; must approve all physical modifications (drilling, cabling).	Access to deck plans, technical specifications of the ship, and installation "windows of opportunity."	Supportive but Protective: Cautious about disruptions to cruise schedules or damage to ship interiors.	Monthly executive summaries, coordination meetings.
Mystic Invest Holding (COO / Sponsor)	Ensure the project adds value to the fleet and stays within the corporate financial framework.	High: Has ultimate veto power; can cancel the project, change the budget, or re-prioritize resources.	Financial backing, final approval of the project charter, and high-level strategic direction.	Supportive but Cost-Conscious. Risk: Might cut funding if labor or equipment costs escalate.	High-level monthly milestone reports, budget vs. actual tracking.
Radio Engineering Dept. (Internal Tech)	Ensure network architecture is sound, frequencies optimized, and 100% coverage across all passenger decks.	High: Technical studies determine the number of base stations. Errors lead to "dead zones" or dropped calls.	Coverage heat-maps, frequency planning, equipment specs, and final technical validation.	Professional and Technical. Risk: May request expensive equipment that conflicts with budget.	Technical coordination meetings, sharing of CAD deck plans.

Continued on next page

Stakeholder (name and role)	Stakeholder's interest in the project?	Stakeholder's potential impact	What does the project expect the Stakeholder to provide?	Perceived attitudes and/or risk	Inputs to Stakeholder Communication Plan
Regulation Dept. (Internal Compliance)	Ensuring the system follows ITU and National Telecommunications Authorities guidelines.	High: If they don't secure the right to use specific frequencies, the system cannot legally broadcast.	Licensing for satellite backhaul, frequency permits, and technical compliance reports.	Compliance-Focused. Risk: Fines or ship detention if frequencies interfere with local networks.	Technical compliance docs, regulatory updates.
Management Dept. (Finance/Admin)	Strict control over project expenditures and optimization of human resource allocation.	High: Can freeze the budget or demand a reduction in team size if ROI decreases.	Approval of the financial plan, authorization for high-cost labor, and budget reconciliation.	Economical and Result-Oriented. Risk: May reject technical expenses exceeding estimates.	Bi-weekly financial reports, labor hour tracking sheets.
Law Dept. (Internal Legal)	Protecting the operator from legal liability and ensuring roaming contracts are enforceable.	High: Must sign off on all contracts with Atlas Ocean Voyages and satellite providers.	Draft and review of Service Level Agreements (SLAs), roaming contracts, and liability waivers.	Risk-Averse: Thorough but slow. Risk: Legal bottlenecks if contracts misalign with maritime law.	Legal review cycles, email updates on contract status.
Buying Dept. (Internal Procurement)	Staying within the project budget while ensuring high-quality hardware is acquired.	Medium/High: Ability to negotiate prices affects the "Financial Management" evaluation of the project.	Purchase orders for hardware; management of delivery timelines and vendor negotiations.	Cost-Conscious. Risk: Long lead times for specialized marine-grade hardware.	Budget tracking reports, procurement schedules.
Marketing Dept. (Internal Commercial)	Create attractive service packages to maximize revenue and passenger satisfaction.	Medium: Defines the User Experience requirements (e.g. 4K streaming), which dictates technical design limits.	Service definitions, pricing models, branding for the "Ship Portal," and promotional material.	Enthusiastic. Risk: May promise features/speeds that the satellite backhaul cannot support.	Marketing requirement docs, feedback on portal User Interface/User Experience.
Continued on next page					

Stakeholder (name and role)	Stakeholder's interest in the project?	Stakeholder's potential impact	What does the project expect the Stakeholder to provide?	Perceived attitudes and/or risk	Inputs to Stakeholder Communication Plan
Ship Captain (Terje Ulset)	Safety of the vessel and passengers; ensuring no interference with bridge navigation/radar.	High (Operational): Can deny installation work or power-on tests if he perceives a safety risk.	Final on-site authorization for work, access to technical spaces, and coordination of "Quiet Hours."	Pragmatic and Safety-First. Risk: May halt work during bad weather or critical maneuvers.	Daily briefing during installation, direct radio contact.
The Constructor (External)	Complete the physical installation according to contract to maintain reputation with Atlas.	High (Execution): Speed and quality determine the physical "Go-Live". Poor work leads to system failure.	Skilled labor, specialized tools for maritime environments, and "As-Built" documentation.	Task-Oriented. Risk: May take shortcuts in hidden areas or experience access delays.	Detailed blueprints, daily "Tool-box" safety meetings.
Satellite Link Supplier	Provide bandwidth services and maintain long-term contract for maritime data traffic.	Critical (Technical): If the link fails or lacks coverage in polar regions, the project value drops to zero.	High-speed backhaul, low-latency connection, and technical support for tracking antennas.	Service-Oriented. Risk: Signal degradation in extreme weather or polar blind spots.	SLAs, real-time bandwidth dashboards, 24/7 hotline.
Equipment Suppliers (External Vendors)	To sell high-end telecommunications hardware and secure maintenance contracts.	Medium: Faulty hardware or lack of maritime certs causes delays in the commissioning phase.	Marine-grade Wi-Fi 6 access points, cellular small cells, cabling, and technical manuals.	Commercial and Supportive. Risk: Hardware incompatibility with ship's power grid.	Technical data sheets, warranty agreements.
Ship Security (Internal Vessel Dept.)	Maintaining ISPS (International Ship and Port Facility Security) standards and preventing unauthorized access to restricted areas.	Medium: Can restrict access to server rooms or the bridge if clearances aren't provided.	Badges for the installation team, oversight of equipment boarding, and credential verification.	Vigilant and Procedural. Risk: Delays in "Work Permits" if background checks are late.	Security clearance requests, daily access logs.
World Voyager Crew (Rest of Crew)	Reliable communication with families and efficient operation of internal ship apps.	Low/Medium: Feedback during testing is vital for identifying "weak spots" in crew quarters.	Cooperation during hardware installation in service areas and reporting network performance.	Supportive and Eager. Risk: Misuse of bandwidth or interference with work duties.	Internal memos via ship's bulletin, feedback surveys.

Continued on next page

Stakeholder (name and role)	Stakeholder's interest in the project?	Stakeholder's potential impact	What does the project expect the Stakeholder to provide?	Perceived attitudes and/or risk	Inputs to Stakeholder Communication Plan
World Voyager Passengers	High-speed Wi-Fi and 5G roaming to share photos and stay connected with family.	High (Indirect): Satisfaction scores determine commercial success and future investments.	Revenue through service subscriptions and feedback on the user interface/portal.	Demanding and High-Expectation. Risk: Negative reviews if signal drops in polar zones.	Welcome brochures in cabins, digital "How-To" guides.
Regulatory Agencies (e.g., ANACOM)	Spectrum management, ensuring no illegal frequency use, and enforcing Mobile Communications on board Vessels rules.	High (Legal): Can fine the operator or order system shutdown if interference occurs in territorial waters.	Granting of radio licenses, allocation of frequency bands, and safety certifications.	Neutral and Strict. Risk: Surprise port inspections and long lead times for approvals.	Formal license applications, EMI/EMC studies.
Port Authorities (External/Local)	Safety of port operations and ensuring no interference with port communications (VHF/Radar).	Medium: Can restrict use of on-board equipment while docked if it interferes with shore networks.	Permission to maintain satellite tracking while docked and adherence to "Radio Silence" zones.	Neutral and Regulatory. Risk: Conflicting local regulations in different international ports.	Port entry notifications, technical declarations.
Other Mobile Operators	Protection of licensed spectrum and potential roaming revenue from passengers.	Medium: Can file complaints with regulators if ship signal "leaks" and interferes with their customers.	Reciprocal roaming agreements and technical parameters for cell handovers in port.	Competitive and Protective. Risk: Legal disputes over spectrum "pollution" or interference.	Roaming contract negotiations, geofencing coordination.

2.1 Stakeholders Power/Interest Matrix

Based on the 19 identified stakeholders, the following matrix categorizes each actor into a management quadrant. This strategic mapping ensures that the Project Leader prioritizes "Key Persons" who hold the most influence over the critical path, while maintaining appropriate communication levels for regulatory bodies and end-users.

Table 2: Project Stakeholders Matrix (Power vs. Interest)

Power / Interest	Low Interest	High Interest
High Power	Keep Satisfied <i>(High Influence / Low Daily Interest)</i> • Law Dept. • Regulation Dept. • Management Dept. (Finance/Admin) • Regulatory Agencies (e.g., ANACOM) • Buying Dept. (Procurement)	Manage Closely <i>(Key Players / High Priority)</i> • Project Leader • Atlas Ocean Voyages (Customer) • Mystic Invest Holding (Sponsor) • Radio Engineering Dept. • Satellite Link Supplier • Ship Captain (Terje Ulset)
Low Power	Monitor <i>(Minimal Effort / Passive)</i> • Equipment Suppliers • Ship Security (ISPS) • Other Mobile Operators • Port Authorities	Keep Informed <i>(High Interest / Low Influence)</i> • World Voyager Passengers • World Voyager Crew • Marketing Dept. (UX/UI Requirements) • The Constructor (Execution/Go-Live)

3 Project Requirements

To ensure the successful deployment and operation of the cellular and Wi-Fi infrastructure aboard the MV World Voyager, the project must fulfill a strict set of requirements.

3.1 Functional Requirements

- **Coverage:** The system must provide both mobile cellular communications and Wi-Fi access across the vessel. However, network capacity must be distributed strategically: public spaces experiencing heavy traffic and simultaneous device usage (such as the pool deck and lounges) must be provisioned with high-density coverage. Yet, passenger cabins will be provisioned with standard, baseline coverage sufficient for individual use.
- **Connectivity:** Ensure connectivity of the system with the land via satellite systems.
- **Port Shutdown:** Build a location-based shut-down mechanism that disables the cellular base stations, to ensure that when the cruise is at port, interference with local mobile operators is avoided.

3.2 Non-Functional Requirements

- **Durability:** The external infrastructure (external access points, antennas and cabling) must resist the marine environment, including high salinity, humidity, and constant vessel vibration.
- **Performance:** The network must manage traffic. It must handle high intensity usage near public zones (e.g. swimming pool), without significantly compromising the connection in the private areas.
- **Aesthetic:** Hardware installed in passenger-facing areas must be visually integrated into the ship's interior design to not disrupt the aesthetic of the ship.
- **Safety:** The hardware must not interfere with safety material nor obstruct access to any part of the ship, whether in passenger or crew areas.

3.3 Business Requirements

- **Passenger Experience:** The project must satisfy the modern expectation that people can use their mobile phones and tablets for voice communications and Internet access in any location.
- **Labor Cost:** All human resources must be inserted into one of the three labor-cost categories: €80/hour, €100/hour, and €120/hour.
- **Equipment Budgeting:** The financial budget must take into account all hardware equipment.
- **Installation and Maintenance:** Physical installation and maintenance costs should be considered. A routine that verifies the functioning of the network must be established. The costs of hiring specified technicians should be taken into account.

3.4 Technical Requirements

- **Fault Tolerance:** All access points must be fault-tolerance. A corrupted device must not influence the performance of those nearby.
- **Interference:** The system must not, under any circumstances, interfere with the ship's navigation systems, radar systems and emergency communication.
- **Licensing:** The international operators must be and maintain the proper licenses and respect maritime law directives to legally provide services while the ship is cruising at sea.

4 WBS

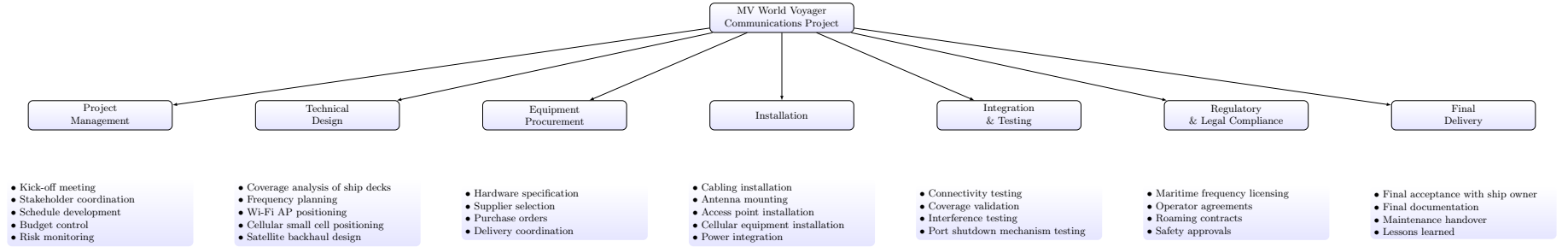


Figure 1: Work Breakdown Structure of the MV World Voyager Communications Project

4.1 Project Management

This section defines the management activities required to coordinate the project. It includes planning, stakeholder alignment, schedule control, financial monitoring, and risk management to ensure that the project progresses according to scope, time, and cost objectives.

Table 3: WBS – Project Management

WBS Work Package	Milestone	Description
Kick-off meeting	Kick-off meeting completed	Formal start of the project involving the main stakeholders. Establishes project objectives, scope, responsibilities, and initial alignment regarding execution strategy and expected deliverables.
Stakeholder coordination	Stakeholder coordination framework established	Continuous coordination of internal and external stakeholders to ensure alignment of technical, operational, legal, and commercial interests throughout project execution.
Schedule development (MS Project)	Baseline schedule approved	Preparation of the detailed project schedule using MS Project, including activity sequencing, duration estimation, dependencies, and identification of critical milestones.
Budget control	Budget monitoring process implemented	Monitoring and control of project financial resources, ensuring that labour, equipment, and operational costs remain aligned with the approved budget.
Risk monitoring	Risk monitoring active	Identification, tracking, and periodic reassessment of project risks to support mitigation actions and minimise impacts on schedule, cost, and technical delivery.

4.2 Technical Design

This section defines the engineering activities required to prepare the technical solution before physical deployment. It includes radio planning, equipment positioning, and communication architecture design to ensure that the installed solution satisfies performance and operational requirements.

Table 4: WBS – Technical Design

WBS Work Package	Milestone	Description
Coverage analysis of ship decks	Coverage analysis approved	Evaluation of signal propagation requirements across all ship areas to determine expected wireless coverage needs and identify high-density usage zones.
Frequency planning	Frequency plan validated	Definition of frequency allocation for onboard communications systems to optimise performance and minimise interference with other radio systems.
Wi-Fi AP positioning	Wi-Fi access point layout approved	Definition of the physical positioning of Wi-Fi access points according to coverage objectives, passenger density, and ship structural constraints.
Cellular small cell positioning	Cellular positioning approved	Planning of the installation locations for cellular small cells to ensure adequate onboard mobile coverage and efficient signal distribution.
Satellite backhaul design	Satellite backhaul design completed	Definition of the communication architecture connecting onboard systems to external networks through satellite links, including bandwidth requirements and integration principles.

4.3 Equipment Procurement

This section covers the activities required to acquire all technical components necessary for project implementation. It includes specification, supplier selection, purchasing processes, and logistics coordination to ensure timely equipment availability.

Table 5: WBS – Equipment Procurement

WBS Work Package	Milestone	Description
Hardware specification	Hardware specifications approved	Definition of all technical equipment requirements, including performance, compatibility, environmental resistance, and installation constraints for maritime use.
Supplier selection	Suppliers selected	Evaluation and selection of suppliers capable of providing compliant telecommunications equipment according to project technical and commercial requirements.
Purchase orders	Purchase orders issued	Preparation and release of formal purchase orders for all required equipment, ensuring alignment with approved procurement plans and budget constraints.
Delivery coordination	Equipment delivered onboard	Coordination of logistics, shipping, and delivery schedules to ensure that all equipment arrives in time for installation activities.

4.4 Installation

This section describes the physical implementation activities required to deploy the telecommunications infrastructure onboard the vessel. It includes the installation of cabling, antennas, access points, cellular equipment, and all related power connections necessary to prepare the system for integration and operational testing.

Table 6: WBS – Installation

WBS Work Package	Milestone	Description
Cabling installation	Cabling installation completed	Installation of all required cabling infrastructure to support the onboard telecommunications system. Includes routing, securing, and terminating cables according to technical design specifications and shipboard installation standards.
Antenna mounting	Antennas mounted and secured	Mounting and physical installation of all antennas in the defined ship locations. Ensures correct positioning, structural fixation, and readiness for connection to the communication network.
Access point installation	Wi-Fi access points installed	Installation of Wi-Fi access points across the ship according to the approved coverage and positioning plan. Ensures proper placement, physical fixation, and readiness for configuration and testing.
Cellular equipment installation	Cellular equipment installed	Deployment and installation of onboard cellular equipment, including small cells and associated hardware. Ensures all devices are mounted, connected, and prepared for integration with the wider communications system.
Power integration	Power integration completed	Integration of all installed telecommunications equipment with the ship’s power supply. Includes power connection, verification of electrical compatibility, and confirmation that all components receive stable and reliable power.

4.5 Integration & Testing

This section covers the activities required to verify that the installed telecommunications solution operates correctly, delivers the expected coverage, and complies with performance and safety requirements. It ensures that the integrated system is technically validated before final acceptance.

Table 7: WBS – Integration and Testing

WBS Work Package	Milestone	Description
Connectivity testing	Connectivity tests successfully completed	Execution of connectivity tests to verify that all installed systems, devices, and network components communicate correctly. Confirms operational links between onboard equipment, internal network infrastructure, and external communication channels.
Coverage validation	Coverage validation approved	Validation of wireless coverage across the ship to confirm that Wi-Fi and cellular services meet the expected performance and design requirements. Includes verification of signal quality, service availability, and coverage consistency in the defined areas.
Interference testing	Interference testing completed	Assessment of possible radio-frequency interference between installed communication systems and other onboard equipment. Ensures that all systems can operate together without unacceptable degradation of performance or disruption.
Port shutdown mechanism testing	Port shutdown mechanism verified	Testing of the port shutdown mechanism to confirm that network ports can be disabled or controlled according to operational and security requirements. Verifies correct behaviour, reliability, and compliance with system protection policies.

4.6 Regulatory and Legal Compliance

This section defines the regulatory and contractual activities necessary to ensure that the onboard telecommunications system can be legally installed and operated. It includes licensing, operator coordination, roaming arrangements, and safety-related approvals required for maritime deployment.

Table 8: WBS – Regulatory and Legal Compliance

WBS Work Package	Milestone	Description
Maritime frequency licensing	Frequency licenses obtained	Preparation, submission, and approval of all required maritime frequency licensing documentation. Ensures that the shipboard telecommunications system is legally authorised to operate within the applicable maritime regulatory framework.
Operator agreements	Operator agreements signed	Definition and formalisation of agreements with telecom operators required to support service provision. Establishes the commercial and operational conditions necessary for network access and service continuity.
Roaming contracts	Roaming contracts activated	Negotiation and activation of roaming contracts to ensure service availability across the intended maritime routes and operational areas. Confirms that users can access communications services under the agreed roaming conditions.
Safety approvals	Safety approvals granted	Obtaining the necessary safety approvals for the installation and operation of telecommunications equipment onboard the ship. Ensures compliance with relevant maritime safety standards, technical regulations, and operational requirements.

4.7 Final Delivery

This section includes the final project closure activities, ensuring that the telecommunications solution is formally accepted, fully documented, and handed over for future operation and maintenance. It also captures the lessons learned to support continuous improvement in future maritime connectivity projects.

Table 9: WBS – Final Delivery

WBS Work Package	Milestone	Description
Final acceptance with ship owner	Final acceptance signed off	Formal review and acceptance of the completed telecommunications solution with the ship owner. Confirms that the project deliverables have been completed, validated, and accepted according to the agreed scope and requirements.
Final Documentation	Final documentation delivered	Compilation and delivery of all final project documentation, including technical records, installation details, test results, compliance evidence, and operational references. Ensures that all relevant information is complete and handed over in an organised manner.
Maintenance handover	Maintenance handover completed	Transfer of maintenance responsibility to the designated operations or support team. Includes delivery of relevant documentation, support instructions, and confirmation that the receiving team is prepared to maintain the installed solution.
Lessons learned	Lessons learned documented	Preparation of the lessons learned record capturing key successes, issues, risks, and recommendations identified during the project. Supports knowledge transfer and continuous improvement for future maritime telecommunications projects.